ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. The design brief, closed-loop control and feedback; pulley blocks, gear drives, work and power; two-hand control, displacement-step diagrams and valve symbols; reading a motor control circuit and short-circuit protection; materials, hand-tool safety and condition monitoring.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers.
5. Draw all symbols, circuits and diagrams neatly, using a ruler.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

QUESTION 1

1.1 Explain the term *design brief* and list THREE things that a design brief includes. (5)

1.2 Draw a labelled block diagram of a closed-loop system that keeps the temperature of an oven constant. (6)

1.3 Explain the term *feedback* in a control system. (2)

1.4 Give TWO examples of mechatronic products found in a home. (2)

1.5 Explain the difference between a manual system and an automatic system, and give ONE example of each. (5)

[20]

QUESTION 2

2.1 A pulley block with a velocity ratio of 4 lifts a load of 800 N with an effort of 250 N. Calculate the following:

2.1.1 The mechanical advantage (2)

2.1.2 The efficiency of the pulley block (3)

2.2 FIGURE 1 below shows a motor driving a machine through two gears.

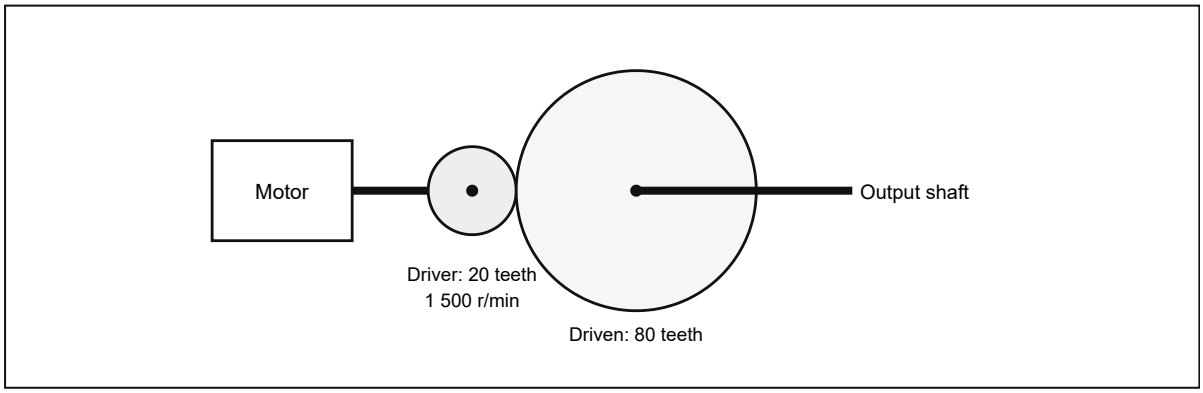


FIGURE 1

2.2.1 Calculate the gear ratio. (2)

2.2.2 Calculate the speed of the output shaft. (2)

2.2.3 The motor delivers a torque of 2 N·m. Calculate the torque at the output shaft if there are no losses. (2)

2.3 A worker pushes a trolley with a force of 50 N over a distance of 20 m in 10 seconds. Calculate the following:

2.3.1 The work done (2)

2.3.2 The power (2)

2.4 Name THREE types of gears. (3)

2.5 State the function of a flywheel. (2)

[20]

QUESTION 3

3.1 FIGURE 2 below shows the pneumatic circuit of a small press.

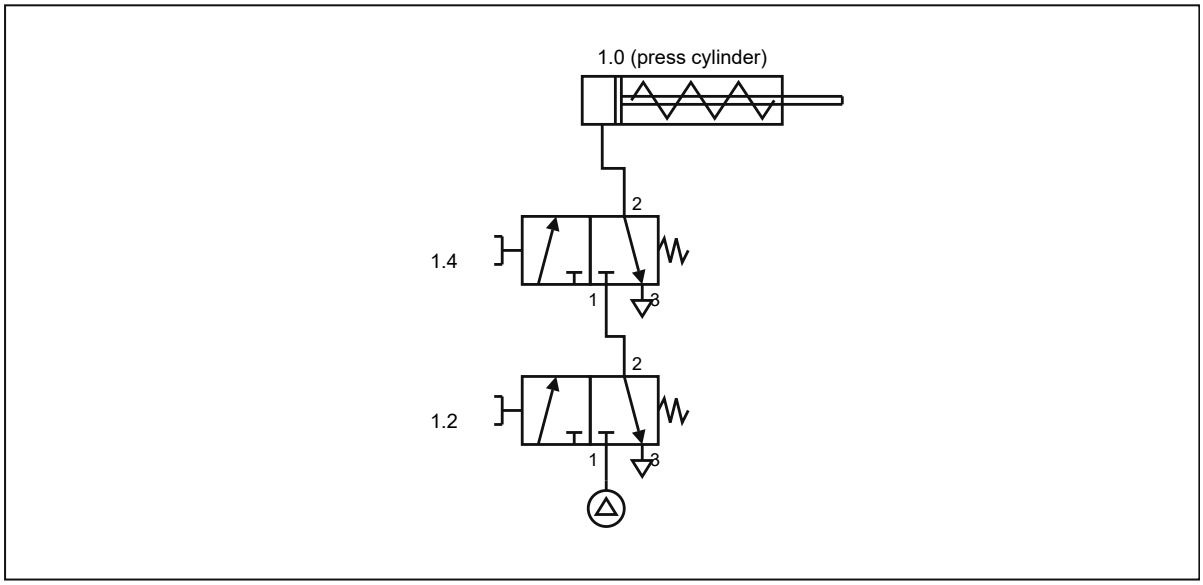


FIGURE 2

3.1.1 Name the logic function formed by valves 1.2 and 1.4. (1)

3.1.2 Explain why this circuit is used on a press. (2)

3.1.3 Name the single valve that can be used instead of connecting valves 1.2 and 1.4 in series. (1)

3.2 Two double-acting cylinders, A and B, must work in the sequence A+ B+ B− A−. Draw the displacement-step diagram for this sequence. (6)

3.3 Draw the symbol of a 5/2-way valve that is operated by a solenoid and has a spring return. (4)

3.4 Explain how a roller-lever valve (limit valve) is used to detect the end position of a cylinder. (3)

3.5 Explain why silencers are fitted to the exhaust ports of pneumatic valves. (3)

[20]

QUESTION 4

4.1 FIGURE 3 below shows a motor control circuit.

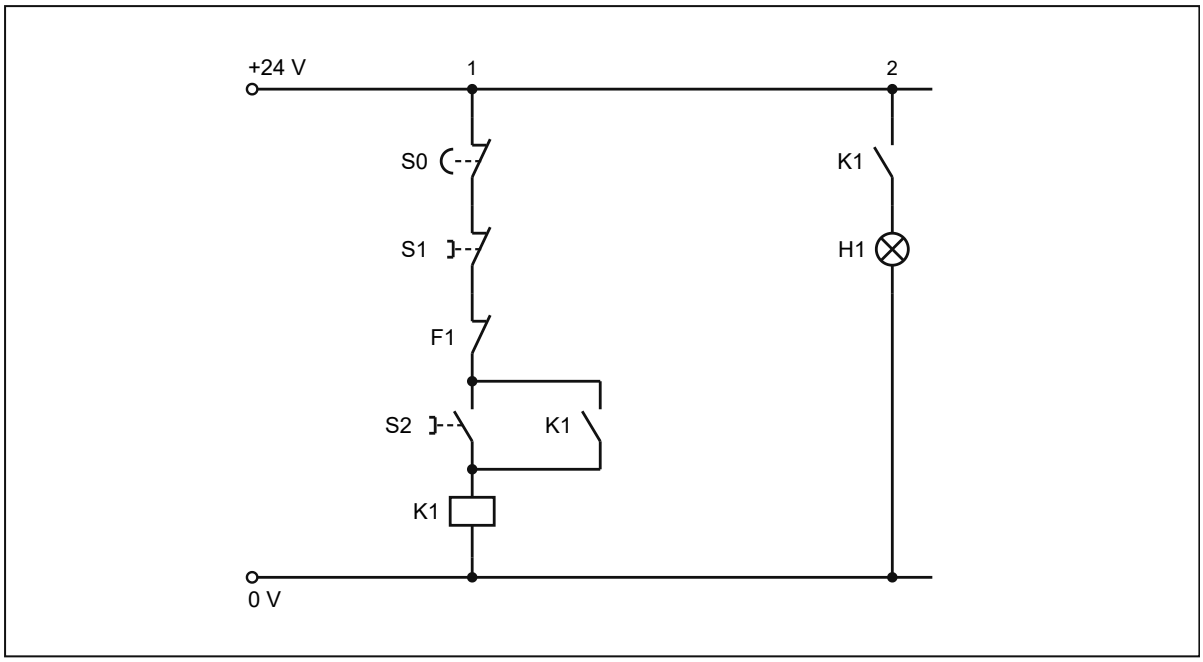


FIGURE 3

4.1.1 Identify the components S0, S1, S2, F1 and K1. (5)

4.1.2 Explain what happens when F1 trips while the motor is running. (2)

4.1.3 What does lamp H1 show? (1)

4.2 Draw the symbol of EACH of the following:

4.2.1 Fuse (2)

4.2.2 Indicator lamp (2)

4.2.3 DC motor (2)

4.3 Explain what a short circuit is, and how a circuit is protected against it. (4)

4.4 Give TWO reasons why the metal frames of machines must be earthed. (2)

[20]

QUESTION 5

5.1 Name TWO ferrous metals and TWO non-ferrous metals used in mechatronic systems. (4)

5.2 Give ONE property and ONE use of EACH of the following materials:

5.2.1 Copper (2)

5.2.2 Aluminium (2)

5.2.3 Nylon (2)

5.3 List FOUR rules for the safe use of hand tools. (4)

5.4 Explain the term *condition monitoring* and give TWO examples of it. (6)

[20]

TOTAL: 100

END OF QUESTION PAPER